Forecast Uncertainty from Bootstrapped Portfolio and Market Returns

Deuel Lau

Abstract

1 Introduction

Extrapolating last year is tempting: it is the most recent evidence, it is easy to compute, and a good year feels like it says something about the next one. The honest question is how much a single year (or four) of daily returns can tell us about next year's return, and how wide the range of plausible outcomes should be.

A common way to put a range around a forecast is the *bootstrap*: resample past daily returns at random, with replacement, to simulate many possible futures, and read the range off those simulations. The method is simple and makes few assumptions, but it treats the past average return as if it were the true one. With only a few years of data that average is itself uncertain, so the range may be too narrow. A range is only useful if it is honest, which we measure by *coverage*: how often the realised outcome falls inside a band with a stated level (a 90% band should contain about 90% of outcomes).

We study this for the daily returns of a single portfolio and of market benchmarks over about four years. We (i) set out how a bootstrap range behaves when uncertainty in the mean is ignored, and compare it with variants that account for it, including a Bayesian *posterior* for the mean (a belief about the average return after combining a prior with the data); (ii) test the coverage of each method on simulated data where the true process is known, including fat-tailed and volatility-clustered returns; and (iii) apply the methods to the real series and report how wide the ranges are and how much a few years of data say about the average return.

This is an illustration of forecast uncertainty. We do not predict returns, rank methods on real data, or claim a trading edge.

2 Data and setup

The data are daily returns for four series from 22 September 2022 to 13 October 2026 (1,018 trading days): the time-weighted return of one personal brokerage account, and three exchange-traded funds used as market benchmarks, SPY (S&P 500), BND (US investment-grade bonds) and GLD (gold). The benchmark returns are total returns, so dividends and interest are included. All series are expressed in Singapore dollars: the benchmarks are listed in US dollars and are converted with the daily USD/SGD exchange rate. Days on which US markets were closed are merged into the next trading day, where the account's returns over the closed days are compounded and the benchmarks' exchange-rate moves are applied. The first and last years are partial: 2022 starts in September and 2026 ends in October. The first account observation is the base date and is dropped.

We work with daily log returns $r_t = \ln(1 + R_t)$, where R_t is the simple daily return, so that returns over several days add. Results are shown as simple returns where this helps interpretation. A forecast made at an origin T uses data up to T only, and the outcome it is scored against is the return over the next H trading days of the same series, so that no future information enters any forecast.

Table 1 reports the number of trading days and the annualised volatility of each series by calendar year. In each full year (2023 to 2025) the account's volatility is higher than that of every benchmark. The raw data are personal and are not released.

Table 1: Trading days and annualised volatility of daily log returns by calendar year. 2022 and 2026 are partial years.

Year	Days	Account	SPY	BND	GLD
2022	70	0.36	0.25	0.11	0.18
2023	250	0.27	0.14	0.08	0.14
2024	252	0.31	0.13	0.07	0.16
2025	250	0.29	0.20	0.06	0.21
2026	196	0.22	0.14	0.06	0.31
All	1018	0.28	0.16	0.07	0.21

3 Methods and theory

3.1 Setup and notation

At a forecast origin we observe a history $r = (r_1, \dots, r_n)$ of daily log returns, with sample mean and plug-in variance

$$\hat{\mu} = \frac{1}{n}\sum_{t=1}^{n} r_t, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{t=1}^{n}(r_t - \hat{\mu})^2.$$

The *empirical distribution* of the history is $\hat{F}_n = \frac{1}{n}\sum_{t=1}^{n}\delta_{r_t}$, which puts probability $1/n$ on each observed day. Drawing a value from $\hat{F}_n$ means picking one of the n observed days uniformly at random, and drawing m values *i.i.d. from* $\hat{F}_n$ means drawing m days with replacement. The bootstrap replaces the unknown distribution of daily returns by $\hat{F}_n$ and simulates from it.

The target is the return over the next H trading days,

$$S = \sum_{t=n+1}^{n+H} r_t.$$

A forecast method returns B simulated values $S_1^*, \dots, S_B^*$ of S. The *band* at level $1-\alpha$ is $[\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}]$, where $\hat{q}_p$ is the empirical p-quantile of $S_1^*, \dots, S_B^*$. Its *coverage* is the probability that the realised S falls inside it, and a band is honest when its coverage is close to $1-\alpha$.

Assumption 1 (i.i.d. returns)**.** *Daily log returns are independent and identically distributed with mean μ and finite variance σ^2, and future returns are independent of the history.*

3.2 Two bootstrap methods

Simple bootstrap. For each path $b = 1, \dots, B$, draw $R_{b,1}^*, \dots, R_{b,H}^*$ independently from $\hat{F}_n$, that is, H days with replacement from the history, and set

$$S_b^* = \sum_{h=1}^{H} R_{b,h}^*.$$

The band at level $1-\alpha$ is $[\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}]$, the empirical quantiles of $S_1^*, \dots, S_B^*$. This treats the historical mean as the true mean.

Mean-uncertainty bootstrap. For each path $b = 1, \dots, B$ there are two stages. In the first, draw a pseudo-history $r_{b,1}^*, \dots, r_{b,n}^*$ independently from $\hat{F}_n$, that is, n days with replacement from the history, and let $\hat{F}_b^* = \frac{1}{n}\sum_{t=1}^{n}\delta_{r_{b,t}^*}$ be its empirical distribution. In the second, draw $R_{b,1}^*, \dots, R_{b,H}^*$

independently from $\hat{F}_b^*$ and set

$$S_b^* = \sum_{h=1}^{H} R_{b,h}^*.$$

The band is again formed from the empirical quantiles of $S_1^*, \ldots, S_B^*$. Every path has its own pseudo-history with mean $\hat{\mu}_b^* = \frac{1}{n}\sum_t r_{b,t}^*$, which differs from $\hat{\mu}$ by about $\hat{\sigma}/\sqrt{n}$, so the simulated sums carry the uncertainty of the sample mean as well as day-to-day noise.

For a fixed history, the simulated sums of the two methods have

$$\begin{aligned} \mathbb{E}^*[S_b^*] &= H\hat{\mu} && \text{(both methods)}, \\ \mathrm{Var}^*(S_b^*) &= H\hat{\sigma}^2 && \text{(simple)}, \\ \mathrm{Var}^*(S_b^*) &= H\hat{\sigma}^2\Big(1 + \frac{H-1}{n}\Big) && \text{(mean-uncertainty)}, \end{aligned}$$

where $\mathbb{E}^*$ and Var^* are taken over the resampling for a fixed history (Appendix A). The pseudo-history has exactly n days, the length of the real history, so that its mean is as uncertain as the real sample mean and no more. Both methods use the same horizon H and the same number of paths B, which affects only Monte Carlo noise, not the spread of the band.

3.3 Coverage of the simple bootstrap

Proposition 1. *Under Assumption 1, the forecast error $S - H\hat{\mu}$ has variance $H\sigma^2(1 + H/n)$, while the simple bootstrap's sums have variance $H\hat{\sigma}^2 \approx H\sigma^2$. If S is approximately normal, a simple-bootstrap band at level $1 - \alpha$ therefore has coverage approximately*

$$2\,\Phi\left(\frac{z_{1-\alpha/2}}{\sqrt{1 + H/n}}\right) - 1,$$

where Φ is the standard normal distribution function and $z_{1-\alpha/2}$ its $1 - \alpha/2$ quantile. The mean-uncertainty bootstrap has variance $H\hat{\sigma}^2\big(1 + (H-1)/n\big) \approx H\sigma^2(1 + H/n)$ and so removes this shortfall.

The proof is in Appendix A. The correction factor $\sqrt{1 + H/n}$ depends only on the ratio of the forecast horizon to the length of the history. For example, with $H = n$ a nominal 95% band covers about 83% of outcomes, and a nominal 80% band about 64%.

4 Simulation study: i.i.d. returns

4.1 Design

We simulate independent Normal daily log returns with an annual volatility of 28% (about that of the account) and an annual mean of 8%. For each history length $n \in \{250, 500, 750, 1000\}$ days and horizon $H \in \{21, 63, 126, 252\}$ days, each of 2,000 independent replications simulates one path of $n + H$ days. The first n days are the history and the outcome is the sum of the last H, so the outcome is the continuation of the same path and never an independent draw. Each method forms bands at the 80% and 95% levels from 2,000 simulated paths, and the *coverage* of a cell is the share of replications whose outcome falls inside the band. Its standard error is $\sqrt{p(1-p)/R}$ for R replications, between 0.5 and 1.1 percentage points here.

All settings were fixed before any simulation was run, together with three predictions: (a) the simple bootstrap's coverage matches Proposition 1 to within 2 percentage points in every cell; (b) cells with the same H/n lie on one curve whatever n is; (c) the mean-uncertainty bootstrap is within 2 percentage points of its nominal level in every cell.

4.2 Results

Figure 1 shows coverage against H/n and Table 2 lists every cell. The simple bootstrap's coverage falls as H/n grows and follows the curve of Proposition 1. When H/n is close to one ($n = 250$,

$H = 252$), a nominal 95% band contains the outcome in only 82% of replications and a nominal 80% band in 61%. Points from different history lengths lie on the same curve (prediction b).

Prediction (a) holds only approximately. Of the 32 readings (16 cells at two levels), 29 are within two standard errors of the theoretical value, but 3 differ from it by more than 2 percentage points, the largest by 2.4. All three are at $H = 252$, and two of them come from the same cell and so share the same replications. On average the simple bootstrap's coverage is 0.3 percentage points below the theoretical value.

The mean-uncertainty bootstrap stays close to its nominal level in every cell (prediction c): the largest gap is 1.7 percentage points, and 31 of the 32 readings are within two standard errors. Its bands are wider than the simple bootstrap's by a factor that matches $\sqrt{1 + H/n}$ to within 0.005.

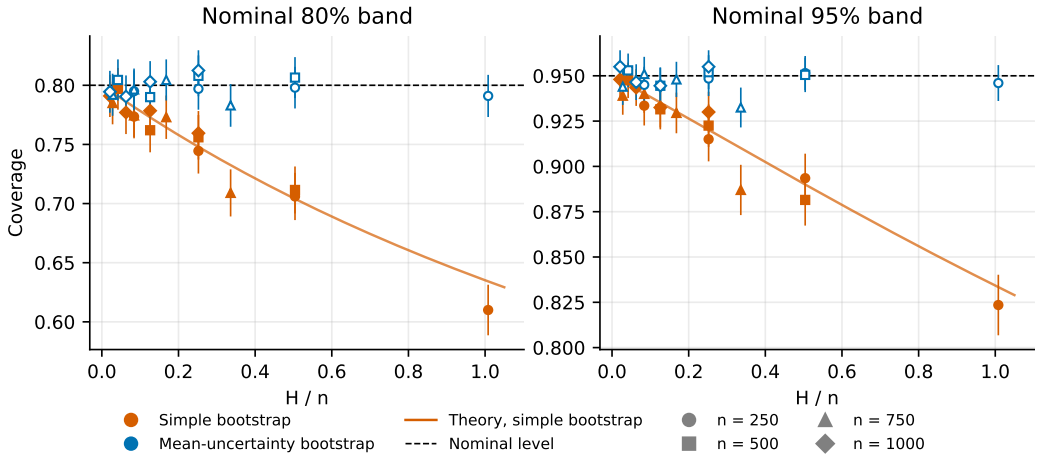


Figure 1: Coverage of the simple and mean-uncertainty bootstrap bands against H/n in the simulated i.i.d. world, at nominal 80% (left) and 95% (right). Markers show the history length n; bars are 95% intervals from the simulation. The solid line is the prediction of Proposition 1 for the simple bootstrap and the dashed line is the nominal level.

Table 2: Coverage in the simulated i.i.d. world. Theory is Proposition 1; Simple and Mean-unc. are the simple and mean-uncertainty bootstrap.

		95% band			80% band		
n	H	Theory	Simple	Mean-unc.	Theory	Simple	Mean-unc.
250	21	0.940	0.933	0.945	0.782	0.773	0.795
250	63	0.920	0.915	0.949	0.748	0.745	0.797
250	126	0.890	0.893	0.952	0.704	0.706	0.798
250	252	0.833	0.824	0.946	0.634	0.610	0.791
500	21	0.945	0.948	0.953	0.791	0.797	0.804
500	63	0.935	0.931	0.945	0.773	0.762	0.790
500	126	0.920	0.922	0.952	0.748	0.756	0.808
500	252	0.890	0.881	0.951	0.704	0.712	0.806
750	21	0.947	0.939	0.944	0.794	0.785	0.792
750	63	0.940	0.940	0.951	0.782	0.774	0.796
750	126	0.930	0.929	0.948	0.764	0.773	0.804
750	252	0.910	0.887	0.932	0.732	0.709	0.783
1000	21	0.948	0.948	0.955	0.795	0.791	0.794
1000	63	0.943	0.944	0.947	0.786	0.777	0.790
1000	126	0.935	0.932	0.945	0.773	0.778	0.803
1000	252	0.920	0.930	0.955	0.748	0.759	0.812

This is the simplest case: returns are Normal with constant volatility, which favours the theory. It says nothing yet about fat tails or volatility clustering.

A Derivation of Proposition 1

By Assumption 1, S is independent of $\hat{\mu}$, so

$$\mathrm{Var}(S - H\hat{\mu}) = \mathrm{Var}(S) + H^2 \,\mathrm{Var}(\hat{\mu}) = H\sigma^2 + \frac{H^2\sigma^2}{n} = H\sigma^2\Big(1 + \frac{H}{n}\Big).$$

The simple bootstrap resamples the history, so its sums have variance $H\hat{\sigma}^2 \approx H\sigma^2$ and miss the second term. A band at level $1-\alpha$ has half-width $z_{1-\alpha/2}\sqrt{H}\,\sigma$ around $H\hat{\mu}$. If the forecast error is approximately normal with standard deviation $\sqrt{H}\,\sigma\sqrt{1+H/n}$, the probability that it lies inside the band is $2\Phi\big(z_{1-\alpha/2}/\sqrt{1+H/n}\big) - 1$.

For the simple bootstrap, the H draws are i.i.d. from $\hat{F}_n$, which has mean $\hat{\mu}$ and variance $\hat{\sigma}^2$, so $\mathbb{E}^*[S^*] = H\hat{\mu}$ and $\mathrm{Var}^*(S^*) = H\hat{\sigma}^2$.

For the mean-uncertainty bootstrap, let $\hat{\mu}_b^*$ and $\hat{\sigma}_b^{*2}$ be the mean and plug-in variance of the b-th pseudo-history, which is a sample of n i.i.d. draws from $\hat{F}_n$. Then $\mathbb{E}^*[\hat{\mu}_b^*] = \hat{\mu}$, $\mathrm{Var}^*(\hat{\mu}_b^*) = \hat{\sigma}^2/n$ and $\mathbb{E}^*[\hat{\sigma}_b^{*2}] = \hat{\sigma}^2(n-1)/n$. Given the pseudo-history, the H forecast days are i.i.d. from $\hat{F}_b^*$, so S_b^* has conditional mean $H\hat{\mu}_b^*$ and conditional variance $H\hat{\sigma}_b^{*2}$. By the law of total variance,

$$\mathrm{Var}^*(S_b^*) = H\,\mathbb{E}^*[\hat{\sigma}_b^{*2}] + H^2\,\mathrm{Var}^*(\hat{\mu}_b^*) = H\hat{\sigma}^2\frac{n-1}{n} + \frac{H^2\hat{\sigma}^2}{n} = H\hat{\sigma}^2\Big(1 + \frac{H-1}{n}\Big),$$

and $\mathbb{E}^*[S_b^*] = H\,\mathbb{E}^*[\hat{\mu}_b^*] = H\hat{\mu}$. This matches the variance of the forecast error up to a term of order $1/n$.